

Exercise 9: Implementing the Command Pattern

CODE:

```
// Receiver Class

class Light {

    public void turnOn() {

        System.out.println("Light is ON");

    }

    public void turnOff() {

        System.out.println("Light is OFF");

    }

}

// Command Interface

interface Command {

    void execute();

}

// Concrete Command - Turn On

class LightOnCommand implements Command {

    private Light light;

    LightOnCommand(Light light) {

        this.light = light;

    }

    public void execute() {

        light.turnOn();

    }

}

// Concrete Command - Turn Off

class LightOffCommand implements Command {

    private Light light;

    LightOffCommand(Light light) {
```

```

        this.light = light;
    }

    public void execute() {
        light.turnOff();
    }
}

// Invoker Class
class RemoteControl {
    private Command command;

    public void setCommand(Command command) {
        this.command = command;
    }

    public void pressButton() {
        command.execute();
    }
}

// Main Class
public class Main {
    public static void main(String[] args) {
        Light light = new Light();
        Command lightOn = new LightOnCommand(light);
        Command lightOff = new LightOffCommand(light);
        RemoteControl remote = new RemoteControl();
        System.out.println("Pressing ON Button");
        remote.setCommand(lightOn);
        remote.pressButton();
        System.out.println();
        System.out.println("Pressing OFF Button");
        remote.setCommand(lightOff);
    }
}

```

```
remote.pressButton();  
}
```

OUTPUT:

```
Pressing ON Button  
Light is ON  
  
Pressing OFF Button  
Light is OFF
```